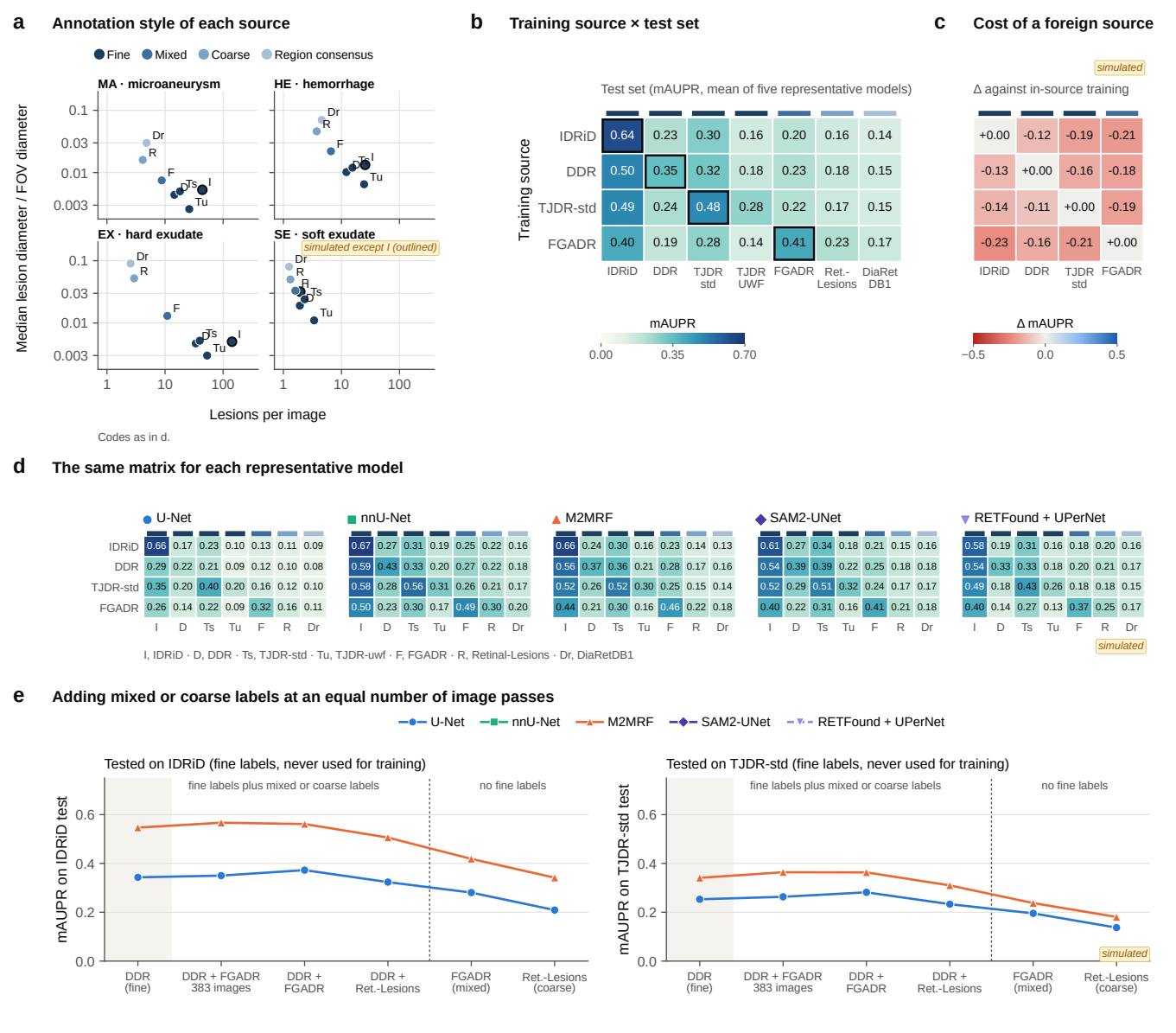

a
Annotation style of each source
Fine
Mixed
Coarse
Region consensus
MA · microaneurysm
HE · hemorrhage
EX · hard exudate
SE · soft exudate
simulated except I (outlined)
Median lesion diameter / FOV diameter
Lesions per image
Codes as in d.
b
Training source × test set
Test set (mAUPR, mean of five representative models)
Training source
IDRiD
DDR
TJDR-std
FGADR
0.64 0.23 0.30 0.16 0.20 0.16 0.14
0.50 0.35 0.32 0.18 0.23 0.18 0.15
0.49 0.24 0.48 0.28 0.22 0.17 0.15
0.40 0.19 0.28 0.14 0.41 0.23 0.17
IDRiD
DDR
TJDR std
TJDR UWF
FGADR
Ret.- Lesions
DiaRet DB1
mAUPR
0.00
0.35
0.70
c
Cost of a foreign source
simulated
Δ against in-source training
+0.00 -0.12 -0.19 -0.21
-0.13 +0.00 -0.16 -0.18
-0.14 -0.11 +0.00 -0.19
-0.23 -0.16 -0.21 +0.00
IDRiD
DDR
TJDR std
FGADR
Δ mAUPR
−0.5
0.0
0.5
d
The same matrix for each representative model
U-Net
nnU-Net
M2MRF
SAM2-UNet
RETFound + UPerNet
IDRiD 0.66 0.17 0.23 0.10 0.13 0.11 0.09 | 0.67 0.27 0.31 0.19 0.25 0.22 0.16 | 0.66 0.24 0.30 0.16 0.23 0.14 0.13 | 0.61 0.27 0.34 0.18 0.21 0.15 0.16 | 0.58 0.19 0.31 0.16 0.18 0.20 0.16
DDR 0.29 0.22 0.21 0.09 0.12 0.10 0.08 | 0.59 0.43 0.33 0.20 0.27 0.22 0.18 | 0.56 0.37 0.36 0.21 0.28 0.17 0.16 | 0.54 0.39 0.39 0.22 0.25 0.18 0.18 | 0.54 0.33 0.33 0.18 0.20 0.21 0.17
TJDR-std 0.35 0.20 0.40 0.20 0.16 0.12 0.10 | 0.58 0.28 0.56 0.31 0.26 0.21 0.17 | 0.52 0.26 0.52 0.30 0.25 0.15 0.14 | 0.52 0.29 0.51 0.32 0.24 0.17 0.17 | 0.49 0.18 0.43 0.26 0.18 0.18 0.15
FGADR 0.26 0.14 0.22 0.09 0.32 0.16 0.11 | 0.50 0.23 0.30 0.17 0.49 0.30 0.20 | 0.44 0.21 0.30 0.16 0.46 0.22 0.18 | 0.40 0.22 0.31 0.16 0.41 0.21 0.18 | 0.40 0.14 0.27 0.13 0.37 0.25 0.17
I D Ts Tu F R Dr
I, IDRiD · D, DDR · Ts, TJDR-std · Tu, TJDR-uwf · F, FGADR · R, Retinal-Lesions · Dr, DiaRetDB1
simulated
e
Adding mixed or coarse labels at an equal number of image passes
U-Net
nnU-Net
M2MRF
SAM2-UNet
RETFound + UPerNet
Tested on IDRiD (fine labels, never used for training)
fine labels plus mixed or coarse labels
no fine labels
mAUPR on IDRiD test
DDR (fine)
DDR + FGADR 383 images
DDR + FGADR
DDR + Ret.-Lesions
FGADR (mixed)
Ret.-Lesions (coarse)
Tested on TJDR-std (fine labels, never used for training)
mAUPR on TJDR-std test
simulated